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Paper 1: Pharmacokinetic Evaluation of a Cinnamon Product on CYP2A6 Substrate Drugs: Appl

Bibliographic Information

- **Title:** Pharmacokinetic Evaluation of a Cinnamon Product on CYP2A6 Substrate Drugs: Application of a Novel Tool Involving the Nicotine Metabolite Ratio
- **Authors:** Nguyen AHP, Hebert DL, Teneralli DA, Wang JR, Lande ME, Chohan NB, Thummel KE, Paine MF
- **Journal:** Clinical Pharmacology & Therapeutics (2025)
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Quick Take

This clinical pharmacokinetic study evaluates the effect of a standardized *Cinnamomum verum* cinnamon product on the pharmacokinetics of two CYP2A6 substrates—nicotine and letrozole—in healthy adults. Despite in vitro predictions of a 4- to 5-fold AUC increase, the cinnamon product showed no clinically relevant interaction, with geometric mean ratios near unity. A novel ‘slope approach’ using the nicotine metabolite ratio provided additional mechanistic insight into CYP2A6 inhibition dynamics.

Executive Summary

This study rigorously tested the hypothesis that a typical dose of *C. verum* (2 g thrice daily) would increase exposure to CYP2A6 substrates due to mechanism-based inhibition by cinnamaldehyde. In a crossover design with 16 healthy volunteers, no significant changes in nicotine or letrozole pharmacokinetics were observed, contradicting earlier in vitro predictions. The work highlights the importance of clinical verification of natural product-drug interactions and introduces a novel analytical tool—the NMR slope approach—to evaluate time-dependent enzyme inhibition. The findings support the safety of *C. verum* consumption with CYP2A6 substrates, but caution that higher doses or other cinnamon species may still pose interaction risks.

Scientific Context & Motivation

CYP2A6 is a human cytochrome P450 enzyme involved in the metabolism of nicotine and letrozole, among others. In vitro studies identified cinnamaldehyde and methoxycinnamaldehyde as mechanism-based inactivators of CYP2A6, predicting substantial drug interactions when cinnamon is co-consumed. However, the clinical relevance remained unknown due to lack of well-controlled trials. This study fills that gap by evaluating a commercially available *C. verum* product at a typical dietary supplement dose, addressing both pharmacokinetic endpoints and developing a new methodology (slope analysis) to capture inhibition dynamics.

✂ Methodological Snapshot

This was an open-label, five-arm crossover study. Participants received single doses of nicotine (2 mg gum) or letrozole (2.5 mg) with and without multiple oral doses of C. verum (2 g thrice daily for 5 days). Plasma concentrations of parent drugs and major metabolites were measured by LC-MS/MS. Non-compartmental analysis (Phoenix WinNonlin) was used to estimate PK parameters. The NMR (3-hydroxycotinine/cotinine) was assessed at 4, 6, and 8 hours and as AUC3-8h. A novel slope approach determined the rate of change of NMR over time.

☒ Structural Model Breakdown

Not applicable as the study used non-compartmental analysis (NCA) rather than compartmental models. The slope approach involves a simple linear regression of NMR vs. time, not a structural model.

Detailed Methodological Analysis

Modeling Approach

Non-compartmental analysis (NCA) to estimate C_{max} , t_{max} , AUC, $t_{1/2}$, CL/F . Geometric mean ratios (GMR) with 90% CI and paired t-tests. Additional slope approach using linear regression.

Data Sources

16 healthy non-nicotine-using adults (18-64 years). Data from 5 arms: Arm 1: single C. verum dose; Arm 2: baseline nicotine; Arm 3: C. verum + nicotine; Arm 4: baseline letrozole; Arm 5: C. verum + letrozole. Plasma samples collected over 12 h (nicotine) or 240 h (letrozole). Detailed in Methods.

Estimation Methods

Non-compartmental analysis via Phoenix WinNonlin (v8.4). For NMR slope: linear regression of NMR vs. time over all time points or a partial linear interval.

Model Evaluation

NCA endpoints evaluated using 90% confidence intervals and pre-specified no-effect boundaries (0.80-1.25). For the slope approach, individual spaghetti plots and comparison of GMRs were used.

Covariate Analysis

None formal; however, potential covariates such as CYP2A6 genetic polymorphisms were discussed as influencing NMR variability but were not assessed.

Statistical Rigor Assessment

Sample size was powered a priori for a 20% change in AUC with 80% power (nicotine) and >90% power (letrozole). Statistical analyses employed paired t-tests on log-transformed endpoints, with a pre-defined no-effect interval of 0.80-1.25. Sensitivity analyses were not explicitly performed, but the inclusion of both AUC and slope endpoints provides robustness. Missing data due to

below-LOQ values were handled by omission, which could introduce bias. The small sample size after exclusions (n=10 for both drugs) limits power to detect modest interactions. The lack of a positive control weakens the ability to conclude the slope analysis is sensitive.

☒ Key Findings

Nicotine AUC_{0-4h} GMR: 0.98 [0.91-1.07]; letrozole AUC_{∞} GMR: 1.10 [0.98-1.24]; both within the pre-defined no-effect range of 0.80-1.25. C_{max} and half-life also near unity. The nicotine metabolite ratio (3-hydroxycotinine/cotinine) at single time points showed no significant change (GMR ~0.90). However, the slope of NMR vs. time over the most linear interval showed a GMR of 0.83 [0.70-0.98], suggesting a potential 17% reduction in the rate of metabolite formation, consistent with modest CYP2A6 inhibition. Individual spaghetti plots revealed considerable inter-individual variability, with some participants showing parallel slopes and others reduced slopes.

🔗 Clinical & Regulatory Implications

The study indicates that C. verum at typical dietary supplement doses (6 g/day) is unlikely to affect the pharmacokinetics of CYP2A6 substrates, suggesting no need for dose adjustments when co-administered with nicotine replacement therapies or letrozole. This fills an important safety knowledge gap for the understudied area of natural product-drug interactions. The slope approach, while not immediately clinically implementable, offers a framework for future interaction studies to assess time-dependent inhibition. Regulatory implications: these data may support labeling that C. verum is safe with CYP2A6 substrates, though caution is warranted for high-dose cinnamon extracts.

Strengths & Limitations

Strengths

- Well-characterized cinnamon product with quantified cinnamaldehyde and methoxycinnamaldehyde content.
- Rigorous crossover design with adequate washout and multiple dosing regimen to assess steady-state inhibition.
- Inclusion of two clinically relevant CYP2A6 substrates (nicotine and letrozole).
- Development of a novel slope approach to capture time-dependent enzyme inhibition dynamics.
- Comprehensive bioanalytical methods with low inter-day variability.

Limitations (Acknowledged by Authors)

- Small sample size and exclusion of some participants due to nicotine use or below LOQ concentrations.
- Inability to directly quantify cinnamaldehyde concentrations in plasma due to instability and low oral bioavailability.
- The slope approach may not be practical for routine clinical phenotyping.

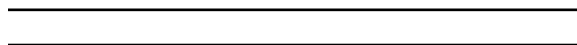
- The absence of a positive control (potent CYP2A6 inhibitor) limits validation of the slope methodology.
- Lack of data on cinnamaldehyde pharmacokinetics and absorption in humans.

Limitations (Expert Review)

- The study used only one specific *C. verum* product; results may not apply to other cinnamon species (e.g., *C. cassia*) with different phytochemical profiles.
- The repeated dosing of cinnamon was only 5 days; longer exposure might lead to different effects.
- Nicotine gum administration may introduce variability due to absorption behavior.
- The slope analysis, while novel, relies on sufficient metabolite formation; for drugs with different half-lives, interpretation may differ.

Generalizability

The findings are generalizable to healthy adults consuming typical doses of *C. verum* (up to 6 g/day) with CYP2A6 substrates. However, generalizability to patient populations with altered hepatic function, those on other medications, or those consuming higher doses of cinnamon or other species is limited. The slope approach may be generalizable to other enzyme-mediated interactions where a metabolite-to-parent ratio can be measured over time.



Figures & Tables

- **Figure 1:** Geometric mean plasma concentration-time profiles for nicotine, cotinine, 3-hydroxycotinine, and letrozole at baseline and after *C. verum* exposure.
 - *Significance:* Visually demonstrates the superimposability of profiles, confirming no major interaction at the population level.
- **Figure 2:** Individual spaghetti plots of AUC and C_{max} for nicotine and letrozole at baseline and after *C. verum* exposure.
 - *Significance:* Shows inter-individual variability and lack of systematic shift, reinforcing the absence of a pharmacokinetic interaction.
- **Figure 3:** Representative nicotine metabolite ratio (NMR) vs. time plots illustrating parallel vs. reduced slope scenarios.
 - *Significance:* Explains the slope approach concept and provides mechanistic scenarios for interpreting time-dependent CYP2A6 inhibition.
- **Figure 4:** Individual spaghetti plots of NMR AUC3-8h and slope values at baseline and after *C. verum* exposure.
 - *Significance:* Illustrates variability in the inhibition effect, with some participants showing reduced slope (possible inhibition) and others not.
- **Table 1:** Pharmacokinetic parameters for nicotine, cotinine, 3-hydroxycotinine, and letrozole at baseline and after *C. verum* exposure, including GMRs and 90% CI.
 - *Significance:* Primary outcome table; all GMRs lie within the pre-defined 0.80-1.25 no-effect interval, confirming lack of interaction.

- **Table 2:** Nicotine metabolite ratio endpoints (NMR at 4, 6, 8 hours; AUC_{3-8h} ; slope) at baseline and after *C. verum* exposure.
 - *Significance:* Shows a marginal reduction in slope (GMR 0.83 [0.70-0.98]), suggesting possible mild time-dependent inhibition that was not captured by traditional endpoints.
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Code & Reproducibility Assessment

The study used Phoenix WinNonlin for NCA, which is proprietary. No custom code or datasets were publicly shared. The analytical methods are described in sufficient detail to allow replication. The slope approach analysis was performed using standard linear regression, presumably in statistical software (not specified).

Supplementary Materials

Supplementary materials include detailed methods for cinnamon product characterization (Table S1), inclusion/exclusion criteria (Table S2), LC-MS/MS conditions (Table S3), study design timeline (Figure S2), and individual NMR vs. time plots (Figure S3). These support the main findings and provide additional granularity.

Future Directions

Future studies should consider using a higher dose of cinnamaldehyde (e.g., via purified cinnamaldehyde capsules) to achieve measurable plasma concentrations and confirm the slope approach with a positive CYP2A6 inhibitor like methoxsalen. Larger sample sizes and inclusion of CYP2A6 genotyping could clarify the contribution of genetic polymorphisms to the observed variability in NMR. A physiologically based pharmacokinetic (PBPK) model of cinnamaldehyde absorption and disposition would help reconcile in vitro and in vivo predictions. Additionally, studies with other cinnamon species (e.g., *C. cassia*) are warranted given differences in coumarin and cinnamaldehyde content.

Expert Commentary

This study exemplifies the critical gap between in vitro mechanism-based inhibition predictions and clinical reality. The disconnect observed here—where a predicted 5-fold AUC increase translates to no interaction—is a common theme in natural product-drug interaction research, often due to poor oral bioavailability, extensive first-pass metabolism, and inadequate hepatic exposure of the inhibitor. The slope approach is a clever addition that leverages the time course of the metabolite ratio to extract dynamic information, but its value is limited by the small sample and absence of a positive control. The authors are commended for their rigorous characterization of the cinnamon product, which allows for quantitative comparison across studies. For pharmacometricians, this is a reminder that in vitro-in vivo extrapolation (IVIVE) of enzyme inhibition requires robust estimates of in vivo inhibitor concentrations at the enzyme site; without such data, predictions can be misleading.

Bottom Line

This well-conducted clinical study provides strong evidence that a standard dose of C. verum cinnamon does not cause clinically relevant pharmacokinetic interactions with CYP2A6 substrates such as nicotine or letrozole, contradicting earlier in vitro predictions. The novel NMR slope approach offers a refined tool for assessing time-dependent enzyme inhibition in interaction studies, though its practical utility beyond research remains limited. Pharmacometricians should note the disconnect between in vitro and in vivo findings, reinforcing the critical role of clinical PK studies in natural product-drug interaction evaluation.

☒ Figures

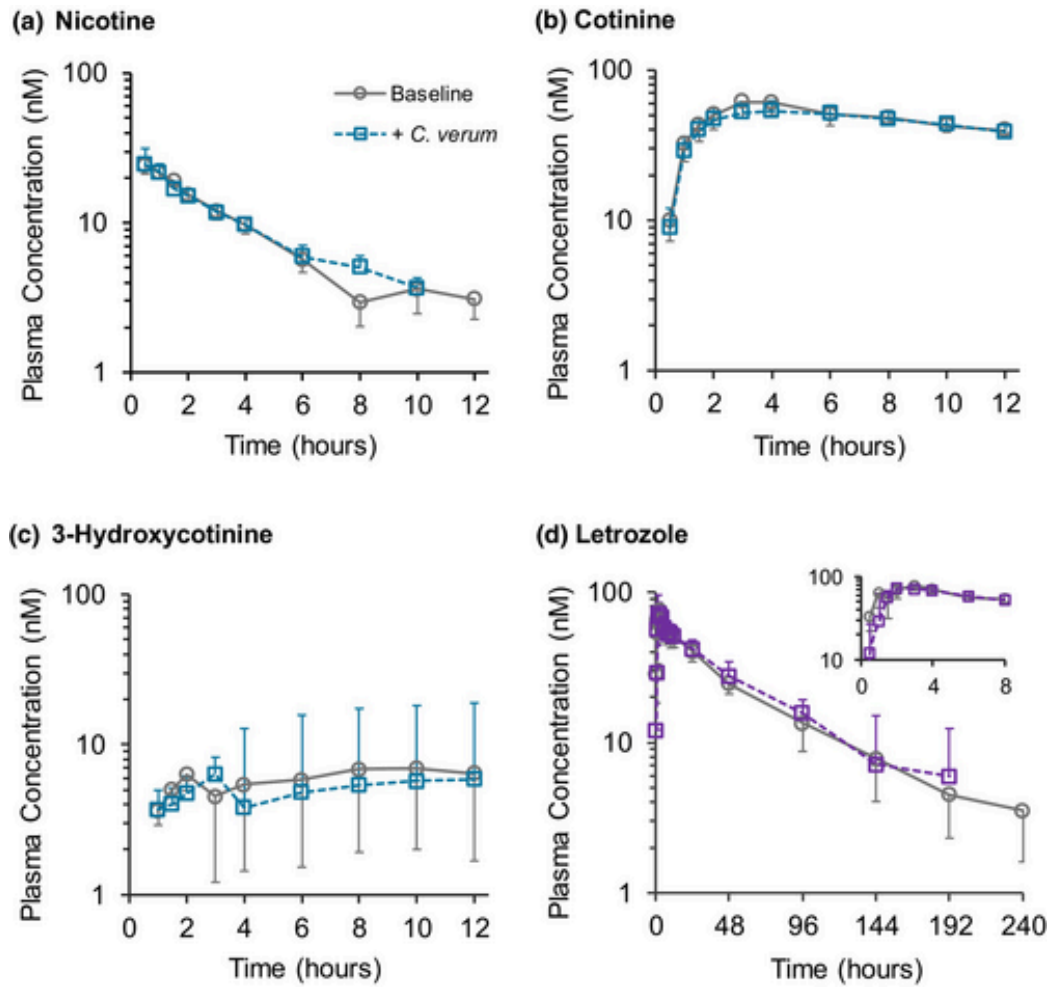


Figure 1: Geometric mean plasma concentration vs. time profiles for (a) nicotine (n= 10), (b) cotinine (n= 11), (c) 3-hydroxycotinine (n= 11), and (d) letrozole (n= 10) fo

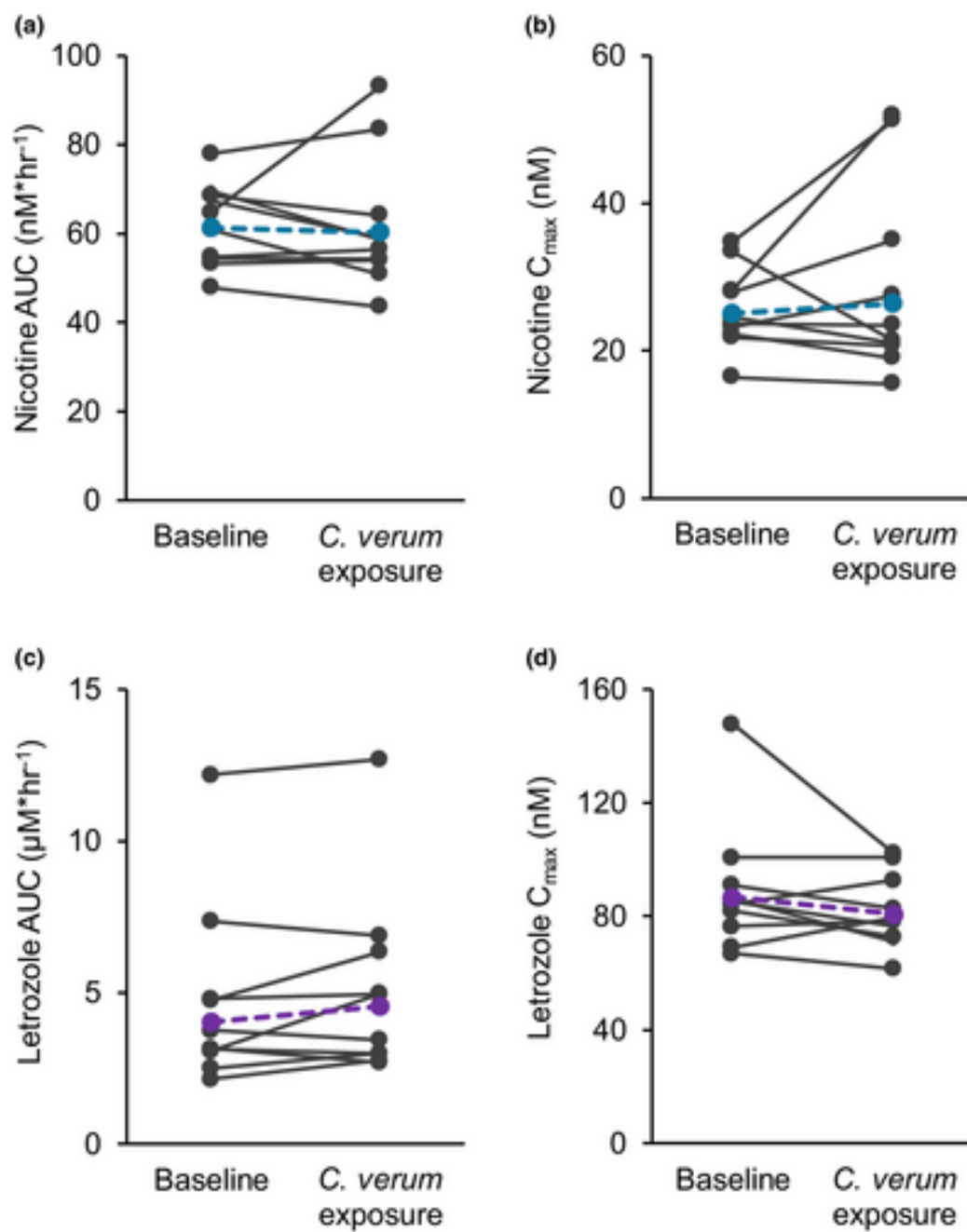


Figure 2: Individual nicotine (a) AUC (n= 10) and (b) C_{max} (n= 10) and letrozole (c) AUC (n= 10) and (d) C_{max} (n= 10) at baseline and after exposure to *C. verum* (2 g) thrice dai

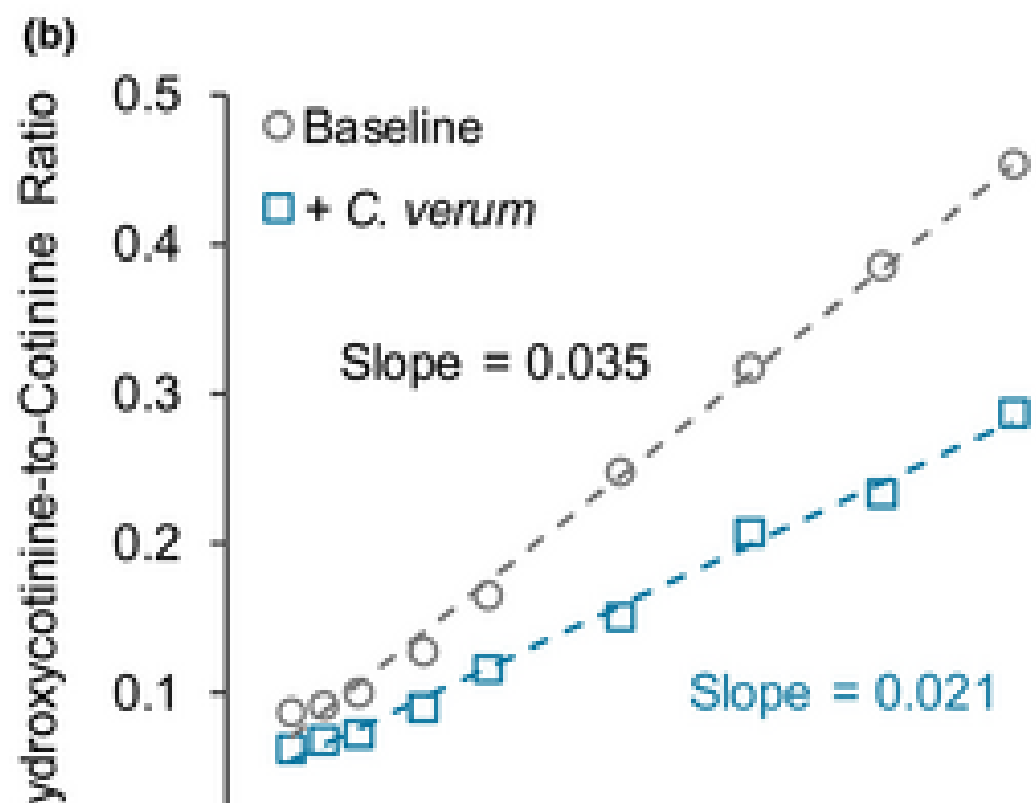
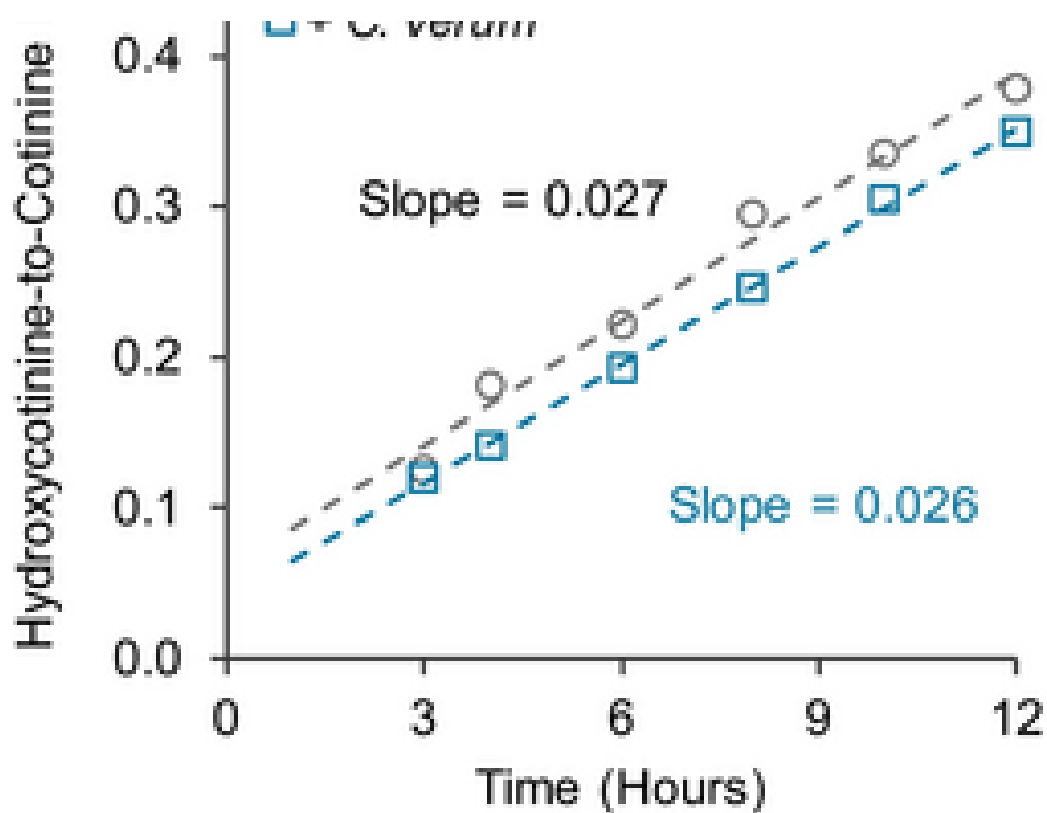


Figure 3: Representative nicotine metabolite ratio vs. time plots showing the rate of metabolite formation (slope) at baseline and after exposure to *C. verum*. Panel a shows n

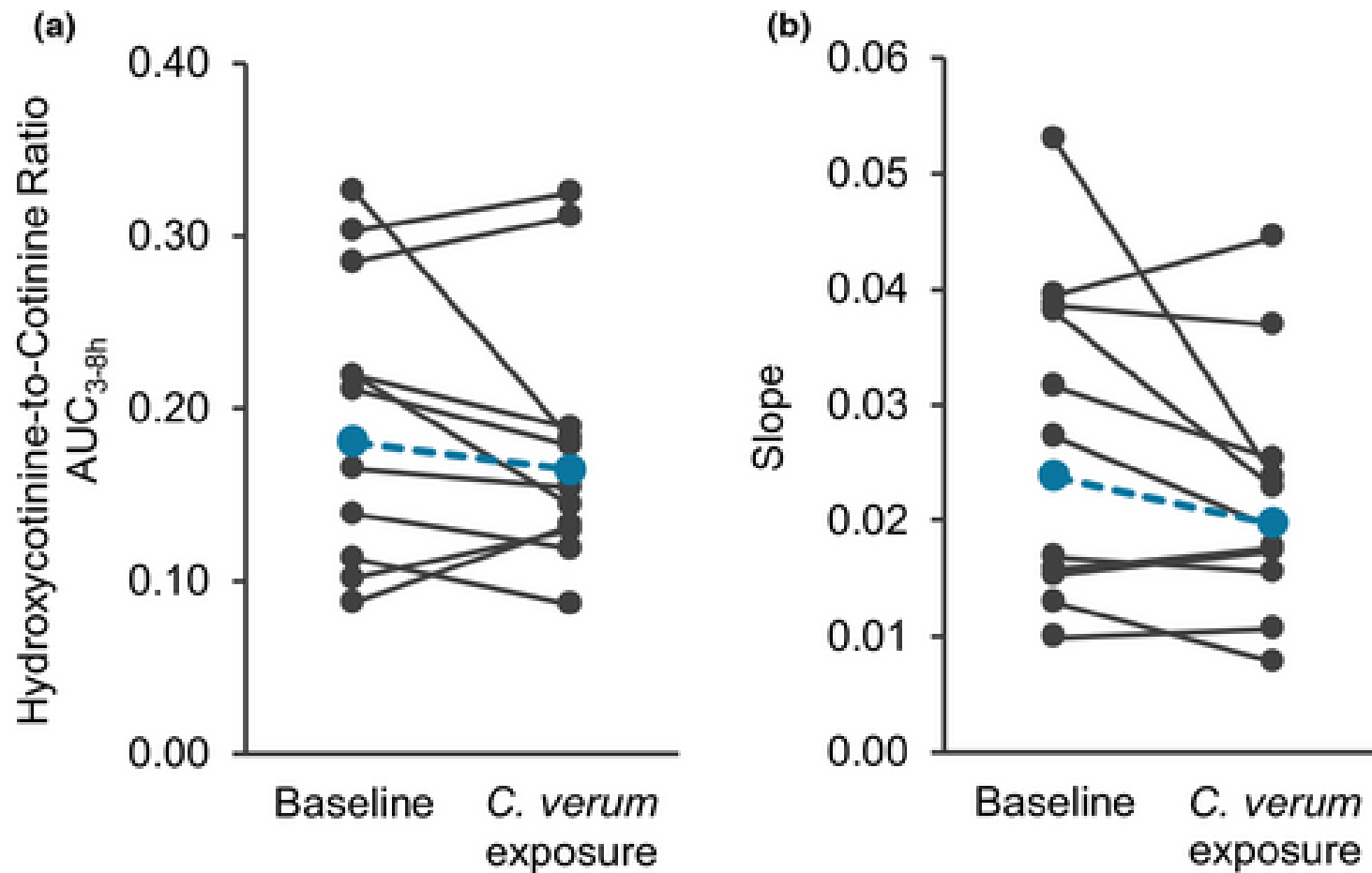


Figure 4: Individual nicotine metabolite ratio AUC_{3-8h} (left, $n=11$) and slope of nicotine metabolite ratio vs. time (right, $n=11$) at baseline and after exposure to *C. verum*.